Varun Patil

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Education

International Institute of Information Technology, Hyderabad — Hyderabad, TG
Master of Technology in Product Design and Management — *Aug. 2025 – Present*

JSPM's Rajarshi Shahu College of Engineering — Pune, MH
Bachelor of Technology in Computer Science Engineering — *Aug. 2020 – June 2024*

Experience

Software Engineering Intern — Pune, MH
Data Axle — *Jan. 2024 – July 2024*

- Developed and maintained backend services and web applications using **C#, ASP.NET Core, SQL Server, and React**.
- Built REST APIs and data-processing workflows, improving application performance through query optimization and efficient database design to support customer acquisition, business intelligence, and data management platforms.
- Improved software delivery through **CI/CD enhancements**, automated testing, and **Azure DevOps-based** deployment pipelines.

Projects

Atelier | *WebSockets, CRDTs, Distributed Systems, PostgreSQL, Prometheus* — GitHub

- Built a real-time collaborative code editor supporting **20+ concurrent users** with **sub-50 ms synchronization latency** using CRDTs (Yjs) based synchronization, distributed WebSocket gateways, and offline conflict resolution.
- Designed an **offline-first collaboration** model enabling local edits to be merged automatically after reconnection.
- Implemented **live presence tracking** and cursor synchronization to provide a shared editing experience.
- Implemented remote code execution by spawning isolated pseudo-terminals (node-pty) on the server, piping stdin/stdout over WebSockets to an xterm.js terminal, supporting JavaScript, Python, C++, and Java
- Engineered a time-travel replay player that rebuilds a room's exact state at any moment in history by folding a recorded CRDT event log – reconstructing the document (in a read-only Monaco editor), who was present, changes to the editor.
- Architected a **5-agent autonomous AI team (Planner/Coder/Reviewer/Tester/Debugger)** as first-class CRDT participants with event-sourced runs, retrieval-grounded prompts via a live tree-sitter code graph, and a human-in-the-loop approval edits with reasoning visible in real time and in replay.

Transformer LMs | *Python, PyTorch, SentencePiece, NumPy, Apple MPS, Fine-Tuning, Web Scraping* — GitHub

- Implemented two independent 25.3M-parameter, 12-layer decoder-only Transformers with causal multi-head self-attention, pre-LayerNorm residual blocks, GELU feed-forward networks, learned positional embeddings, and tied input/output embeddings.
- Built corpus pipelines processing 1.18B training tokens using Unicode NFC normalization, SHA-256 exact deduplication, deterministic train/validation/test splits, and separate 10K-vocabulary SentencePiece BPE tokenizers with byte fallback.
- Pretrained each model on 500M tokens using AdamW, gradient accumulation and clipping, linear warmup, cosine learning-rate decay, and memory-mapped token streams; implemented deterministic checkpoint recovery restoring optimizer, scheduler, RNG, and data-stream state.
- Applied answer-only supervised finetuning to synthetic comparative and transitive reasoning tasks, increasing candidate exact-match accuracy from 30.33% to 70.67% for Hindi and 31.11% to 63.61% for Nepali across 1,800 held-out examples per language.

Judgment Explained — Quantized LLM Inference & RAG Verification System | *Python, PyTorch, HF Transformers, pydantic*

- Engineered a document-grounded RAG pipeline for Indian judgments selected from the 7,130-pair IN-Abs corpus, combining PDF/text ingestion, BM25 passage retrieval, fixed evidence pools, and paragraph-level citation provenance.
- Implemented a tool-calling structured LLM verification workflow with schema-validated tool calls, bounded retries, evidence-access controls, and source-versus-claim speaker/status checks to detect and repair attribution errors.
- Designed reproducible experiment infrastructure using immutable snapshots, SHA-256 integrity checks, isolated inference sessions, and execution traces; developed automated tests for provenance, malformed inputs, and tool boundaries.
- Benchmarked three system variants across 16 judgments and 64 controlled claims, measuring attribution accuracy, information coverage, repair success, latency, and token consumption with paired statistical analysis.
- Reduced attribution errors by 6% points versus a generic evidence-checking baseline while retaining 90% key-information coverage.

Technical Skills

Languages Java | Python | C/C++ | C# | JavaScript

Frameworks React | ASP.NET Core | Node.js | Rust | Flask | JUnit | RestAPI | FastAPI

Machine Learning & LLM PyTorch | Deep Learning | HF Transformers | LLM Inference | RAG

Distributed Systems Apache Kafka | Redis | FAISS | Docker | FastAPI | WebSockets | CRDTs

CS Fundamentals OS | DBMS | Computer Networks | Object Oriented Design | Data Structures & Algorithms

Achievements

- Secured **AIR 202** in **PGEE CSE 2025**, qualifying for admission to IIIT Hyderabad.
- Achieved **AIR 2849** in **GATE CS 2025** among 170,000+ candidates, corresponding to the **98.14 percentile**.
- Ranked **9th** overall in the Shaastra Techathon 2024, IIT Madras, competing among 829 entrants